Imad Serhal

Automation & Control Systems Engineer

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PROFILE

Engineer with an M.Sc. in Automation & Control Systems Engineering (University of Padova) and a B.Eng. in Mechanical Engineering. At STIGA I carry out tests on electric and autonomous machines from start to finish: I set up the test benches and instruments, collect and analyse the data-logger and CAN-bus data, and write the reports against standards and internal norms. Before that I worked in mechanical design for residential buildings in Lebanon, where I designed HVAC and pumping systems, prepared Excel calculations and AutoCAD drawings, and supported installation on site. During my studies I completed PLC projects in Siemens TIA Portal with Factory I/O, and I want to grow my career in industrial automation.

TECHNICAL SKILLS

PLC & Automation (university projects): Siemens TIA Portal V15 (IEC 61131-3), S7-PLCSIM, Factory I/O, sequential logic and interlocks, sensor and actuator I/O, analog level control

Communication: CAN bus (J1939, CANopen) log analysis and fault codes, REST APIs

Programming & Scripting: Python (PySide6/Qt, pandas, NumPy, matplotlib), C/C++ (STM32, Arduino), MATLAB scripting

Control Engineering: PID/PI control, cascaded loops, anti-windup, state-space and LQR control, observers, Kalman filtering, system identification, MATLAB/Simulink

Documentation & Office: Technical reports, specifications and engineering drawings, Microsoft Excel (engineering calculation sheets), Word templates, AutoCAD

Hardware: Motors and motor controllers, batteries, sensors and actuators, embedded computers (NVIDIA Jetson), LiDAR

AUTOMATION PROJECTS

PLC automation portfolio (Siemens TIA Portal V15, S7-PLCSIM, Factory I/O) **University projects**

- **Conveyor sorting by height:** PLC program that reads height sensors, routes workpieces to the correct outfeed line and handles operator start/stop from a control panel.
- **Multi-level elevator:** sequential control with floor requests, position sensing and direction interlocks, handing loads to the conveyor at each level.
- **Tank water-level control:** process control from an analog level transmitter, driving fill and discharge valves to hold the tank in its operating band, with an operator panel.
- Each program was verified against a simulated 3D plant through S7-PLCSIM (virtual commissioning). Demo videos at imadhserhal.github.io.

M.Sc. Thesis: Control of an Electrically Excited Synchronous Motor (IVECO heavy-duty EV) **2024**

- Designed the full control architecture in MATLAB/Simulink (FOC, MTPA, flux weakening, MTPV) with lookup tables and the switching logic between operating regions. Extremum Seeking Control cut simulated Joule losses by up to 10%.

Embedded real-time control (STM32) and GPS waypoint robot (Arduino) **University projects**

- Bare-metal C firmware for PWM, encoders, PI speed control and a line follower. GPS navigation with a calibrated magnetometer, Bluetooth commands and ultrasonic sensing.

PROFESSIONAL EXPERIENCE

Control & Testing Engineer **Apr 2025 - Present**

STIGA Italy

- Carried out 25+ tests from start to finish: set up the test benches, measurement setups and instruments, collected and analysed the data, and wrote and issued the reports for R&D project teams, each with procedure, acceptance criteria, results and a pass/fail verdict.
- Wrote the functional validation checklists (based on EN 50636) for an autonomous robot's firmware and app, and used them to run acceptance checks of firmware releases (production SOP and over-the-air updates) on Android and iOS. Compared firmware versions on power consumption and performance.

- Verified electrical and electronic components against standards and internal norms: PCB homologation including vibration tests, DC-DC converter thermal and protection tests (OCP, OVP, UVP), battery charging and insertion life cycles, emergency-stop function and brakes to ISO 5395-3.
- Measured voltage, current, power and temperature with Hioki data loggers and read motor speed, bus current and fault codes from CAN-bus logs to find root causes and support firmware improvements.
- Integrate and debug motors, motor controllers, batteries, sensors, LiDAR and NVIDIA Jetson computing units on prototype machines.
- Built a Python desktop tool on my own initiative that turns data-logger and CAN-bus logs into a finished Word test report. It replaced manual work in spreadsheets and connects to the team's task system through a REST API.

Research Trainee, Advanced Motor Control **Feb 2024 - Jun 2024**

University of Padova Italy

- Built linear and nonlinear control-oriented models of IPM, SPM and EESM electric machines and tuned cascaded current and speed loops with PI control and anti-windup.

Mechanical Design Engineer, Building Services (MEP) **Jun 2021 - Sep 2022**

My Projects SAL Lebanon

- Sized pumps, fans and VRV compressor systems for residential and high-rise projects using Excel calculation sheets (friction losses, static pressure, heat loss).
- Produced AutoCAD shop drawings for HVAC, water supply and drainage, and prepared the engineering documentation used for installation.
- Supported installation activities on customer sites.

EDUCATION

M.Sc. Automation & Control Systems Engineering **2022 - 2024**

University of Padova, Italy. Courses: Advanced Control, Digital Control, Embedded Systems, Robotics, System Identification

B.Eng. Mechanical Engineering **2017 - 2022**

Beirut Arab University, Lebanon. Courses: Control Theory, System Dynamics, Thermodynamics, Fluid Mechanics

LANGUAGES

English: C1 (fluent) | Arabic: native | Italian: A2 (improving)

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